

Methods for Safely Co-Designing Embodied Mental Health VR Interventions with Youth

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Figure 1: Students are sensitized to emotion regulation and experience a demonstration of the different capabilities of VR. **Left:** body attunement activity. **Center:** participants experiencing VR. **Right:** participants acting out their final presentations.

ABSTRACT

Emotion regulation is a vital skill for youth, yet traditional therapeutic models often overlook the somatic nature of emotional experiences. VR offers a promising solution through immersive embodiment. We suggest designing effective interventions requires involving youth as experts in their lived experiences. This paper presents preliminary methodological insights from a 7-day co-design workshop with 20 drama students (ages 15–18). We reflect on how to cultivate safety, scaffold reflection on sensitive topics, and support embodied ideation for VR-based emotion regulation tools. By integrating drama techniques (e.g., tableau, role-play, and perspective-taking), our workshop demonstrates how embodied co-design captures “tacit knowledge” that verbal or written methods may miss. Findings highlight the importance of established trust, the role of community champions, and the efficacy of drama as a precursor to and vessel for VR design. Ultimately, this approach empowers youth to externalize internal states, providing insights for developing future embodied mental health VR interventions.

Index Terms: Co-design, Adolescents, Methodology, Child-Computer Interaction, Research Methods, Mental Health, Embodiment, Virtual Reality

1 INTRODUCTION

Emotion regulation, the ability to modulate the intensity and duration of emotion states, is a critical mental health skill for youth to develop [6]. However, youth are still developing and refining the skills needed to implement adaptive strategies, and may continue to need external supports such as reminders, instructions, and scaffolded opportunities to practice [4, 12]. Traditional therapeutic models prioritize verbal and written communication, but emotions are inherently somatic [11]. There is a need for youth-centred interventions that incorporate non-verbal, body-based modalities to help them learn to express and process emotions that words cannot fully capture.

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One potential solution is virtual reality (VR), since VR can go beyond traditional digital tools by placing the user physically within the therapeutic environment. Rather than only viewing content, youth can inhabit alternative digital bodies (i.e., avatars) that may change their self-perception and engage their entire sensorimotor system through 3D spatial interaction. This sense of spatial presence and visceral immersion allows for alternative forms of learning that may enhance the felt practice of emotion regulation by amplifying existing supports for emotion regulation skills development or speeding up the learning process in ways that are not possible with other non-digital and non-VR digital supports [3].

While there appears to be great potential in utilizing VR’s body-based modalities to support emotion regulation, it remains an open question how to design VR-supported emotion regulation interventions for developing minds and bodies. Both the Psychology and HCI literature advocate for youth participation in design to give them agency over the technology they will eventually use [2, 9, 10]. We need to work with youth on designing VR-based emotion regulation interventions to not only ensure fit, increase access, and enhance effectiveness [5]; but also because we have an ethical obligation to empower youth to participate in technology development that impacts them [1, 7]. This requires safe ways to facilitate reflection and expression of sensitive topics and emotions with this age group. To that end, we share our preliminary methodological insights into our co-design workshop to explore how researchers can cultivate safety and trust, scaffold reflection of lived experiences, and support embodied design ideation in order to investigate sensitive emotional challenges and co-design VR solutions with youth.

2 METHODOLOGY

We conducted a non-consecutive 7-day workshop with a grade 10–12 drama class (N=20, ages 15–18 years). For context, the drama class runs year-round (September–June), and we conducted our workshops in May/June. Thus, the class had already established ways of working together and a high level of trust by the time we approached them. This trust was established in large part by the drama teacher, who had many years of experience working with high school drama students with different learning abilities and socioeconomic backgrounds. She was able to integrate our research goals with her curriculum by facilitating drama activities that fostered connection to our bodies, emotions, and other people. Her involvement was instrumental to the success of our workshop.

DAY 1: Introductions & Sensitization
• Safe space (student-led code of conduct) • Dramatization guessing game (emotion strategies) • Individual reflection (lived experiences)
DAY 2: Challenges & Sharing
• Body & muscle awareness meditation • Group sharing & discussion (lived experiences)
DAY 3: Performance Work Period
• Drama warm-up activities • Selecting & acting out emotional situation
DAY 4: VR Upskilling
• Video introduction to VR • Experiencing bespoke social VR worlds (demo)
DAY 5: Integrating VR Solutions
• Adding VR as intervention support to performance • Facilitator feedback
DAY 6: Final Performances
• Drama warm-up activities • Performing lived experiences w/VR-supported solutions • Facilitator feedback
DAY 7: Group Debrief
• Drama warm-up activities • Individual interviews • Group reflection

Figure 2: Workshop schedule. Seven days, 80 minutes each for a total of 9.33 hours.

Our research team consisted of a senior-career researcher in child-computer interaction (Antle), an early-career researcher in VR and 7 years' crisis volunteer work (Kitson), and several research assistants (incl. Kenny). We created a facilitation guide and conducted training sessions with the research assistants who helped us with workshop facilitation. To ensure the safety of both participants and researchers in our workshop, our institutional ethics review board and the school district's ethics review board approved this project. Through a series of activities, we employed embodied co-design ideation to capture the lived experiences of youth related to emotion regulation, and imagine how VR could act as a potential solution (see Figure 2).

We collected and analyzed multiple sources of data (video, audio, worksheets, whiteboard notes, observation notes, photographs) to record contextually rich information that reflected adolescents' lived experiences. You can find more details on the workshop outcomes in our paper [8]. In this work-in-progress, we reflect on the *methods* we used throughout the workshop to cultivate safety and trust, to scaffold reflection on emotional experiences, and to support modified design ideation around a VR-supported emotion regulation intervention for youth.

3 METHODOLOGICAL FINDINGS

3.1 Cultivating Safety and Trust

We established a “safe container” for participants to express and reflect on their emotions. First, by establishing a student-led code of conduct that included turn-taking, listening non-judgmentally, mindful awareness of others' feelings, and respect for confidentiality and privacy. We also had facilitators take part in workshop activities, sitting in a circle, and had the teacher lead warm-up exercises to break down barriers and power dynamics between participants and researchers. We also employed techniques to empower participants, such as modelling vulnerability, empathy, and holding space. We saw the results of this effort in youth expression, authenticity, engagement with others, and taking ownership of their lived experiences (leadership).

3.2 Scaffolding Reflection

We helped our young participants articulate complex feelings through different means. First, we upskilled them on emotion regulation strategies through a dramatization guessing game where each group portrayed a strategy and others guessed. We provided multiple modes of sharing emotions and lived experiences that were not only writing, but also made use of oral discussions in small groups. The unique spin of this workshop was the use of drama techniques that focused on lived experience and emotion regulation. Specifically, techniques included point-of-view (POV), tableau, 3rd person narration, dramatic hyperbole (character development), rewind to redo the situation, and re-imagining of scenarios or alternative emotional reactions to their lived experiences.

3.3 Embodied Design Ideation

The performances captured “tacit knowledge” on emotion regulation struggles and VR-supported solutions that youth could show with acting, i.e., through the bodies. In the progression of the workshop, we moved from silent reflection to discussion to enactment. Here, sensorimotor warm-up and body exercises acted as a precursor to ideation. Youth explored possibilities for VR through our demo sessions, e.g., trying on different avatars and rewinding time. Ideation occurred using both method (i.e., re-imagining lived experience) and imaginative acting. Drama techniques were used as insight into emotional needs through the use of metaphors and as a tool for reflection. For example, groups would role-play different perspectives, like the parent or friend.

4 METHODOLOGICAL CONSIDERATIONS

Based on our workshop, we suggest the following pointers for researchers using embodied co-design for sensitive topics and VR.

1. **Prioritize Existing Community Leadership:** Conduct co-design within existing safe spaces and find someone to champion your project (like the drama teacher). Before committing to a workshop site or group, it may be a good idea to go into a space to observe with a checklist, e.g., does safety and empowerment exist already?
2. **Body-First Sequencing:** For this workshop to be a success, it was essential that we went into an established safe space and not mess it up—we are HCI researchers and not therapists. To better empower youth to share their emotional experiences, we think that starting with embodied activities and ending with writing might be better, because then they will have something to write about.
3. **Technological “Sandboxing”:** Provide early demos of VR capabilities to provide context where youth can “play” in a design sandbox.
4. **Embodied Expression as Data:** Use drama-based metaphors and role-play as a primary data source. These “non-verbal pathways” are often more directly applicable to VR design than verbal interviews or drawings.

Embodied co-design creates space for youth to express themselves and be seen in a way they couldn't otherwise. In our workshop focusing on sensitive emotional topics, it takes the heat off compared to therapy. Designing *with* rather than *for* youth may not yield the most ground-breaking designs, but it does give them a chance to learn about the technology and provide input on their lived experiences.

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REFERENCES

- [1] A. N. Antle and J. P. Hourcade. Research in child–computer interaction: Provocations and envisioning future directions. *International Journal of Child-Computer Interaction*, 32:100374, 2022. [1](#)
- [2] L. Bowler, K. Wang, I. Lopatovska, and M. Rosin. The meaning of “participation” in co-design with children and youth: Relationships, roles, and interactions. *Proceedings of the Association for Information Science and Technology*, 58(1):13–24, 2021. [1](#)
- [3] D. Colombo, A. Díaz-García, J. Fernandez-Álvarez, and C. Botella. Virtual reality for the enhancement of emotion regulation. *Clinical Psychology & Psychotherapy*, 28(3):519–537, 2021. [1](#)
- [4] K. De France and T. Hollenstein. The development of cognitive reappraisal for regulating emotions. *The Oxford handbook of emotional development*, p. 110, 2022. [1](#)
- [5] A. Druin. The role of children in the design of new technology. *Behaviour and information technology*, 21(1):1–25, 2002. [1](#)
- [6] J. J. Gross. *Handbook of emotion regulation*. Guilford publications, 2013. [1](#)
- [7] N. Iivari and K. Kuutti. Critical design in interaction design and children: impossible, inappropriate or critical imperative? In *Proceedings of the 17th ACM Conference on Interaction Design and Children*, pp. 456–464, 2018. [1](#)
- [8] A. Kitson, A. N. Antle, S. Kenny, A. Adhikari, K. Karthik, A. Cimensel, and M. Chan. ‘i call upon a friend’: Virtual reality-based supports for cognitive reappraisal identified through co-designing with adolescents. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, pp. 1–18, 2024. [2](#)
- [9] H. Thabrew, T. Fleming, S. Hetrick, and S. Merry. Co-design of ehealth interventions with children and young people. *Frontiers in psychiatry*, 9:481, 2018. [1](#)
- [10] D. Tsyatkova and C. Storni. A review of selected methods, techniques and tools in child–computer interaction (cci) developed/adapted to support children’s involvement in technology development. *International Journal of Child-Computer Interaction*, 22:100148, 2019. [1](#)
- [11] D. Westbrook, H. Kennerley, and J. Kirk. *An introduction to cognitive behaviour therapy: Skills and applications*. Sage, 2011. [1](#)
- [12] C. J. Willner, J. D. Hoffmann, C. S. Bailey, A. P. Harrison, B. Garcia, Z. J. Ng, C. Cipriano, and M. A. Brackett. The development of cognitive reappraisal from early childhood through adolescence: A systematic review and methodological recommendations. *Frontiers in Psychology*, 13:3412, 2022. [1](#)